

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

1	OVERVIEW	3
2	PURPOSE AND SCOPE	3
3	HELPFUL TERMS	4
4	ASSISTANCE	4
5	CONSIDERATIONS BEFORE CREATING A REDCAP ECONSENT PROJECT?	5
6	WHEN SHOULD YOU CREATE A REDCAP ECONSENT PROJECT?	5
7	REDCAP PROJECT CREATION	6
7.1	OVERALL PROJECT SETTINGS	6
7.1.1	Project titles	6
7.1.2	“Project Setup” tab	6
7.1.2.1	Main Project Settings	6
7.1.2.2	Enable optional modules and customizations	6
7.2	CREATE “PRE_CONSENT” INSTRUMENT	7
7.2.1	Change first instrument’s name	7
7.2.2	Create “Pre_Consent” fields	7
7.2.2.1	Creating/Updating Action Tags	9
7.3	ADDITIONAL CUSTOMIZATIONS	10
7.3.1	Set a Custom Record Label	11
7.3.2	Enable the File Version for “File Upload” fields?	11
7.3.3	Require a ‘reason’ when making changes to existing records?	11
7.4	PREPARE IRB-APPROVED CONSENT PDF FILE FOR UPLOADING TO ECONSENT REDCAP PROJECT	11
7.4.1	Convert each individual ICF page PDF file to image file	12
7.5	CREATE ICF_SURVEY INSTRUMENT	13
7.5.1	Create “ICF_Survey” instrument & enable as survey	13
7.5.2	Create Descriptive Text fields and upload ICF image files	13
7.5.3	Create remaining ICF_Survey fields	15
7.5.4	ICF_Survey survey settings	17
7.6	CREATE ATTEST_SURVEY INSTRUMENT	18
7.6.1	Create “Attest_Survey” instrument & enable as survey	18
7.6.2	Create Attest_Survey fields	18
7.6.3	Attest_Survey survey settings	20
7.7	CREATE PDF_UPLOADS INSTRUMENT	20
7.7.1	Create PDF_Uploads fields	20
7.7.2	Update “ICF_Survey” & “Attest_Survey” Survey Settings for upload to PDF_Uploads	22
7.8	SET UP “ALERTS & NOTIFICATIONS”	22
7.9	ADDITIONAL PROJECT SETTINGS	23
7.9.1	User Rights/Roles	23
7.10	TESTING AND MOVE TO PRODUCTION STATUS	24
8	ECONSENT WORKFLOWS	24
8.1	RECOMMENDED ECONSENT WORKFLOWS USING THIS REDCAP TEMPLATE	24
8.2	REDCAP’S SEND-IT	25
8.2.1	Using Send-It: providing signed informed consent PDFs	25
8.2.2	Using Send-It: participant receipt of signed PDF copies	28
8.2.3	Using Send-It: confirmation that participant downloaded files	29
9	MODIFIED INFORMED CONSENT VERSIONS	30
9.1	TAKE PROJECT OFFLINE & COMMUNICATE WITH STUDY TEAM	30
9.2	MAKE BACKUP COPY OF PROJECT	30
9.2.1	Steps	30
9.3	MAKE COPY OF THE PROJECT TO COMPLETE & TEST REVISIONS	30
9.3.1	Steps	31

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

9.3.2	Preparing the revised IRB-approved modified ICF version for uploading	31
9.3.3	"Pre_Consent" instrument changes	31
9.3.4	"ICF_Survey" instrument changes	32
9.3.5	"Attest_Survey" instrument changes	32
9.3.6	Test revisions in the copy project	32
9.4	REVISE THE "LIVE" ECONSENT PROJECT	32
10	RECONSENTING	33
11	APPENDICES	33
11.1	"FULL RIGHTS" USER ROLE PERMISSIONS	33
11.2	"ATTESTATION" USER ROLE PERMISSIONS	34
12	REVISION HISTORY	36

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

1 Overview

The Penn Medicine (PSOM/UPHS) research community has increasingly expressed a need to complete and document the informed consent process electronically (*hereafter, “eConsent”*).

Penn Medicine REDCap (*hereafter, “REDCap”*) is a HIPAA-compliant EDC system with functionality to complete eConsent workflows and store signed eConsents, under the below restrictions:

- **The research project DOES NOT require FDA oversight/reporting**
 - If any of the below applies to your research project/study, eConsent signatures require compliance with 21 CFR Part 11:
 - IND or IND-exempt
 - IDE or IDE-exempt or an abbreviated IDE
- The study protocol and informed consent document (“ICF”) documents the eConsent process; described in [Section 6](#)
- The IRB has reviewed and approved:
 - The eConsent process (*as documented in the protocol*)
 - The informed consent document; please note that IRB-approval may/may not come with an IRB-stamped ICF document:
 - **Exempt research:** the IRB does NOT provide a stamped ICF document
 - **Non-exempt research:** IRB DOES provide a stamped ICF document

HIGH-LEVEL SUMMARY

This document provides guidance for:

- 1) When to create a REDCap eConsent project
- 2) How to create a REDCap eConsent project
- 3) How to prepare the IRB-approved informed consent document (“ICF”) PDF file for incorporation into the project
- 4) What REDCap instruments to create
 - a) “Pre-consent” data entry form to capture participant- and informed consent-specific data
 - b) Survey for the IRB-approved informed consent document (“ICF”)
 - c) Study Representative Attestation survey
 - d) “File_Upload” instrument used to store PDFs of the signed copies of the above surveys
- 5) Incorporating new ICF versions into the REDCap eConsent project
- 6) How to manage re-consenting participants
- 7) Best practices for ensuring that all necessary data is captured, stored, and shared in a HIPAA-compliant fashion
- 8) eConsent process workflows
 - a) Modifiable to fit individual study requirements
 - b) Using REDCap’s “Send-It” feature to provide signed PDF copies to participants
 - c) Location of the project’s File Repository storing the signed survey PDF files

eConsent projects should **ONLY** be created in the **Penn Medicine REDCap instance** (<https://redcap.med.upenn.edu/>).

REDCap eConsent projects should be created by **REDCap users with a firm understanding of building REDCap instruments**. This guidance discusses the “what to create,” but **MAY NOT** cover the “how to create.”

2 Purpose and Scope

The Office of Clinical Research Operations Unit (“OCR Operations”) has created this guidance to provide a standard process for creating HIPAA-compliant eConsent surveys, workflows, and documentation for the Penn Medicine research community.

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

3 Helpful Terms

- **Action Tag** – functionality that performs a specific action on a field when instrument loads (e.g., “@DEFAULT” → @DEFAULT=’Tom’ on a First Name field automatically enters the value “Tom” in the First Name field when the instrument loads on the screen)
- **Branching Logic** – logic that can be employed when fields in the database need to be hidden based on data entered in preceding fields
- **Choice Label** – multiple-choice fields only; contains the text of choice options seen on instruments
- **Choice Value** – multiple-choice fields only; the raw data value assigned to choice options; while letters can be used, these values should be numbers
- **Data Dictionary** – a CSV spreadsheet containing the data entry fields in your project
- **Data Entry Form** – an instrument that can only be completed by a project user only; requires login to REDCap
- **eConsent Framework** – surveys only; function specific for eConsents; requires survey participants to complete an additional step before their surveys can be marked “Complete” as a way of certifying their identity; automatically stores a PDF of the survey in the project’s File Repository; additional participant information captured by the project, such as date of birth, can be inserted into the stored PDF footers to provide extra documentation of their identity
- **Field** – a single data point on an instrument that captures data
- **Field Label** – the text of the field’s/variable’s question seen on instruments
- **Instrument** – a data entry form or survey created for capturing data; made up of fields
- **Online Designer** – allows you to create/modify/delete instruments and fields
- **Participant** – a person who completes a survey; cannot see other project data
- **Project** – an individual REDCap database; contains the instruments that capture data; all instruments within a project are tied together
- **Record ID** – a unique identifier for each record in your database
- **Record Status Dashboard** – a table that lists all existing records and their status for every instrument
- **Survey** – an instrument that can be completed by someone who is not a project user; accessed by a URL; do not require login to REDCap
- **User** – a person who is given access to a REDCap project (e.g., can login to REDCap and enter the project); project creators are automatically made a user on that project; users must be added to specific REDCap projects to enter instrument data
- **Validation** – restricts the type of data accepted by a field (e.g., a “date” field can only accept a date), available on Text Box fields only
- **Variable** – the name given to an instrument’s field; used to identify field in reports and exports

4 Assistance

To see all functionality described in this guidance, refer to the REDCap project template “eConsent Template - OCR Operations” (PID 34895), provided with the Penn Medicine REDCap project templates.

This guidance discusses the creation of an eConsent project and workflow that will work for most research studies but is not intended to be “one size fits all.”

If your research study requires the capture of data or a workflow not covered in this guidance (i.e., *consents must be completed by a medical physician*), contact OCR Operations for further assistance (psom-ocrops@pobox.upenn.edu - Subject: “REDCap eConsent assistance”).

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

5 Considerations before creating a REDCap eConsent project?

This guidance is specific to creating a REDCap eConsent project to be used for: **1) NEW research studies only** (i.e., prior to participant enrollment), and **2) a stand-alone project** (i.e., research study data is captured in a separate REDCap project).

Why only for NEW research studies?

While it's possible to implement a REDCap eConsent project for actively enrolling studies, it isn't recommended due to the additional effort required to track a study's informed consent documentation in two places (i.e., *X number of participants provided consent on paper while Y number was done electronically*).

Why can't I create a REDCap project that captures both of my study's eConsents AND study data?

For study teams using REDCap to capture research study data, **it's recommended to use a separate eConsent REDCap project** for the below reasons:

- Revisions to informed consent documents are common
 - Revising eConsent REDCap projects is easier in projects separated from study data
- Easier reconsenting: a stand-alone eConsent project allows you to create more than 1 record for each participant that is reconsented
 - Projects capturing participant data are usually limited to a single project record
- These issues are amplified in projects using longitudinal data collection and/or repeated events/instruments

The eConsent instruments can be added to existing study REDCap projects but is not described in this guidance.

Below are additional considerations before creating an eConsent project:

- Study protocols and informed consent forms should specify that that an eConsent process will be used to consent participants and how it will be conducted.
- Identify the study personnel who will be responsible for creating/revising the eConsent project; this should be documented on the study's Delegation of Authority (DoA) log.
- Since research study eConsent projects should only be created in Penn Medicine REDCap, study personnel require an active Penn Medicine REDCap account.

6 When should you create a REDCap eConsent project?

OCR Operations recommends creating a REDCap eConsent project as soon as the study's informed consent document is finalized but **PRIOR** to initial IRB submission. **The eConsent project should remain in DEVELOPMENT Status until the IRB has approved the study's eConsent process. This allows easier incorporation of project and/or workflow changes required by the IRB's review. Once all needed revisions are completed, and the project is thoroughly tested, the project can be moved to PRODUCTION Status.**

Study protocols and informed consent forms should specify that that an eConsent process will be used to consent participants and how it will be conducted. Below is information that should be provided in IRB submissions.

- Planned workflow(s) for completing the eConsent process
 - Confirmation that the Penn Medicine REDCap will be used
 - Will copies of the ICF document be sent for participant review ahead of the consenting study visit? If so, how?
 - Will all participants be consented via eConsent? Or will there be a hybrid approach (i.e., some via eConsent and some in-person)?
 - *Note that an eConsent project can be set up to allow in-person consenting, providing a centralized storage location for all consents, regardless of how the consent was collected.*
 - What method will be used for the eConsent process? Over the phone? Using video conferencing tools? If so, which one?
 - Confirmation that this method complies with Penn Medicine guidance

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

- How will the study team confirm the identity of participants prior to consents?
- PDF examples of the REDCap ICF and Study Attestation surveys
- Planned workflow for providing signed ICF copies to study participants
 - *Note: this guidance provides a secure method to complete this using REDCap's Send-It function ([Section 8.2](#))*
- Where will signed ICF copies be stored? Remain in the REDCap project or downloaded to an appropriate storage medium?
 - How will they be made available for audits/monitoring?
 - Will all documentation be printed?
 - Will the study team provide access to the REDCap eConsent project (*requires auditors/monitors have an active Penn Medicine REDCap account*)?

This documentation should be stored with study files and used for training.

7 REDCap project creation

The below sections describe the required instruments, functionality, and settings needed to create an eConsent REDCap project.

7.1 Overall project settings

7.1.1 Project titles

Use concise, informative, standard formats when naming eConsent projects. They must allow study team members to easily identify the needed study, especially when they work on multiple REDCap projects. OCR Operations recommends that all eConsent project titles include IRB number and PI last name. An example format: "IRB#_PI-Last-Name_Short-Study-Identifier" → "888888_Doe_StudyDemonstrateEconsents."

7.1.2 "Project Setup" tab

7.1.2.1 Main Project Settings

In the **MAIN PROJECT SETTINGS** section, make sure surveys are enabled.

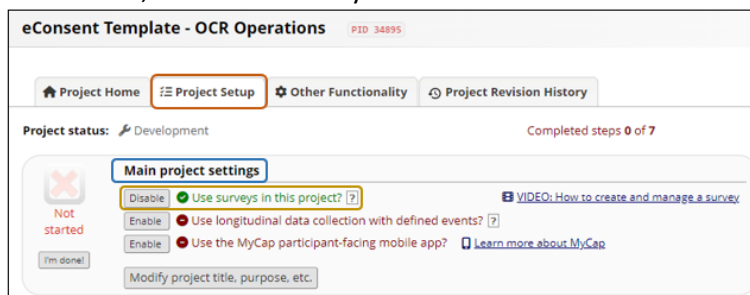


Figure 1. Project Setup tab --> Enable surveys (Project-level)

7.1.2.2 Enable optional modules and customizations

Keep "Auto-numbering for records" ENABLED. **This should be enabled by default.** This ensures that each record created is a unique number.

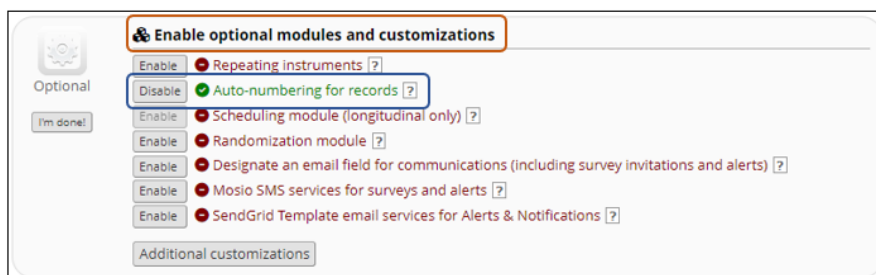


Figure 2. Enable optional modules and customizations --> Keep "Auto-number for records" ENABLED

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

7.2 Create “Pre_Consent” instrument

The project’s first instrument must be a data entry form used to capture participant, and ICF-specific, data (e.g., consent version). Its data will be used to send survey invitations to participants to complete ICF surveys as well as documenting their informed consent process.

7.2.1 Change first instrument’s name

Upon creation of a new REDCap project, the instrument is created automatically; change name to “Pre_Consent.”

7.2.2 Create “Pre_Consent” fields

Create the fields in the below table. Variable names are not required to match; however, following data management best practices is recommended (*i.e., informative – what data the field captures; concise – no more than 26 characters, all lower case, underscores separating elements, etc.*). Using all validation listed in the table is highly recommended.

NOTE: the Record ID field will be created by default as the first field; DO NOT TOUCH IT. If needed, a “Study ID” field can be created to connect a participant’s eConsent data with their study data. This guidance does not include this field because participants may not be assigned their Study ID until after enrollment.

Variable Name	Field Type	Field Label Text	Validation	Branching Logic	Action Tags	Notes
record_id	Text Box	Record ID	Identifier = No	None	None	DO NOT REVISE
N/A	Begin New Section	Participant Information	N/A	None	None	Check “Use the Rich Text Editor” Click the “<>” in the menu Copy/paste below HTML code: <div class=“green” style=“text-align: left;”> <h6><u>Participant Information</u></h6> </div>
icf_ppt_name_f	Text Box	PPT First Name:	Required = Yes Identifier = Yes	None	None	
icf_ppt_name_l	Text Box	PPT Last Name:	Required = Yes Identifier = Yes	None	None	
icf_ppt_email	Text Box	PPT Email Address:	Validation = Email Required = Yes Identifier = Yes	None	None	
icf_ppt_email_confirm	Text Box	Confirm PPT Email Address:	Validation = Email Required = Yes Identifier = Yes	None	None	Entering email address twice used to create a pop-up warning to appear if both email fields don’t match; helps ensure the email address is correct
ppt_email_wrong	Descriptive Text	Error Entered email addresses must match. Please correct.	N/A	[icf_ppt_email_confirm]<>” AND ([icf_ppt_email]<>[icf_ppt_email_confirm])	@HIDDEN-PDF	Check “Use the Rich Text Editor” Click the “<>” in the menu Copy/paste below HTML code:: <div class=“red” style=“text-align: center;”> <h3 style=“text-align: center;”>ERROR</h3> <h4 style=“text-align: center;”>Entered email addresses must match. Please correct.</h4> </div>

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

Variable Name	Field Type	Field Label Text	Validation	Branching Logic	Action Tags	Notes
N/A	Begin New Section	Informed Consent Information	N/A	None	None	Check "Use the Rich Text Editor" Click the "<>" in the menu Copy/paste below HTML code: <pre><div class="green" style="text-align: left;"> <h6><u>Informed Consent Information</u></h6> </div></pre>
icf_vers	Multiple Choice – Drop-down List (Single Answer)	ICF Version # (IRB-Approval Period):	Required = Yes	None	@DEFAULT='-'# (Code of current ICF version choice option)	Captures ICF version information on the IRB-approved ICF document (PDF); <u>Choice Options:</u> Create a choice option for each version of the IRB-approved ICF document, examples: 1, Version 1 (01/01/2021 – 12/31/2021) 2, Version 2 (01/01/2022 – 12/31/2022) 3, Version 3 (01/01/2023 – 12/31/2023) If current IRB-approved ICF document does not have an end date, do not include it (<i>i.e., for protocols that do not require annual continuing review</i>)
icf_process	Multiple Choice – Radio Button (Single Answer)	ICF Process:	Required = Yes	None	None	Provides flexibility – electronic consents can be obtained using REDCap, regardless of whether the enrollment visit is done remotely or in-person <u>Choice Options:</u> 1, Remote 2, In-person
icf_init_recons	Multiple Choice – Radio Button (Single Answer)	Initial or Reconsent?	Required = Yes	None	None	Reconsented PPTs are captured in multiple project records <u>Choice Options:</u> 1, Initial consent 2, Reconsent
icf_study_title	Text Box	Study title (as submitted in HS-ERA)	Required = Yes	None	@DEFAULT = 'STUDY TITLE'; @READONLY	Automatically enters study's title in each record; piped into ICF and Attestation Survey Instructions; update template's @DEFAULT tag with actual study title
icf_pi	Text Box	Study Principal Investigator	Required = Yes	None	@DEFAULT = 'Jane Doe, PhD'; @READONLY	Automatically enters PI name in each record; piped into ICF and Attestation Survey Instructions; update template's @DEFAULT tag with actual PI First and Last Name
icf_irb_num	Text Box	Study UPenn IRB Number:	Validation = Integer Required = Yes	None	@DEFAULT = '888888'; @READONLY	Automatically enters study IRB number in each record; piped into ICF and Attestation Survey Instructions; update template's @DEFAULT tag with actual IRB#

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

Variable Name	Field Type	Field Label Text	Validation	Branching Logic	Action Tags	Notes
icf_prec_comp_dte	Text Box	Today's date: HIDDEN FIELD DEFAULTS TO TODAY'S DATE USED FOR MIN VALIDATION FOR PPT SIGNATURE DATE	Validation = Date (M-D-Y) Required = Yes	None	@HIDDEN @TODAY- SERVER	Used to capture date Pre_Consent instrument completed to use as MIN validation limit for PPT Signature date

Additional REDCap functionality can be used to enhance the look and flow of the form (e.g., *Descriptive Text fields to provide instructions and/or warnings, Field Embedding, etc.*).

7.2.2.1 Creating/Updating Action Tags

To create Action Tags:

- 1) Open the "Edit Field" (or "Create Field" if creating a new field)
- 2) Click the "@ Action Tags" button (bottom of box) to open "@ Action Tags" pop-up menu
- 3) Scroll list to desired Action Tag(s)
- 4) Click "Add" button next to desired Action Tag(s)
- 5) Click "Close" button on pop-up menu

Figure 3. Pink Action Tag box is empty; click "@ Action Tags" button

Figure 4. Click "Add" button to select Action Tag; close Action Tag pop-up menu

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

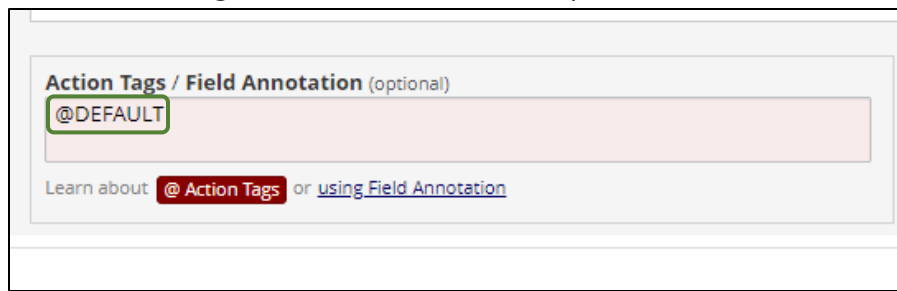


Figure 5. Selected Action Tag inserted

- 6) Some Action Tags (like @DEFAULT) require parameters
 - a) Click anywhere inside pink “Action Tag/Field Annotation” box; opens “Logic Editor” pop-up
 - b) At end of Action Tag, enter:
 - i) Equal sign - =
 - ii) Open parenthesis
 - iii) Opening quote sign (either single or double - ‘ or “)
 - iv) Parameter needed (for @DEFAULT tag for study title, enter the study’s title Test Study: eConsent Template)
 - v) Closing quote sign (same used as opening quote; be consistent throughout the project)
 - vi) Closing parenthesis
 - c) Click “Update & Close Editor” button

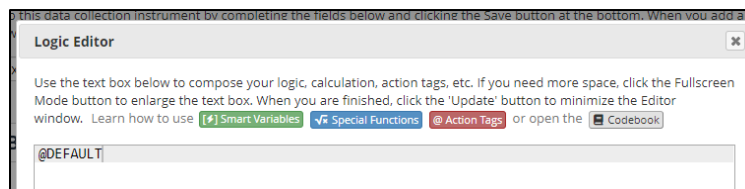


Figure 6. Logic Editor before adding parameters



Figure 7. Logic Editor after parameters; uses @DEFAULT to enter study title and @READONLY to prevent users from changing

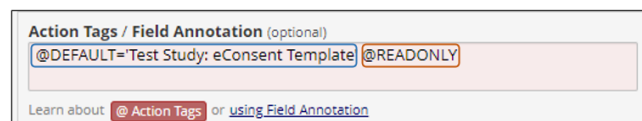


Figure 8. Action Tags/Field Annotation box when finished (Study Title example)

If using the Template REDCap project - Start at Step 6 above and update each field’s Action Tags parameters as needed.

7.3 Additional customizations

Go to the “Project Setup” tab and click the “Additional customizations” button in the “Enable optional modules and customizations” section.

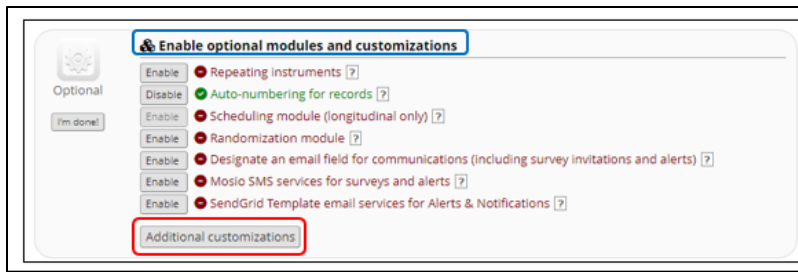


Figure 9. Location of "Additional customizations" button

A pop-up window appears listing customizations that can be applied to the project. Unless specified below, leave all options as their default setting (*i.e. checked or unchecked*). Apply the below customizations.

7.3.1 Set a Custom Record Label

Check the box and create the below custom record label to make record identification easier. This label “pipes” in each participant’s entered name (*last name, first name*); it appears next to the participant’s Record ID everywhere the Record ID is shown (*e.g., Record Status Dashboard, etc.*).

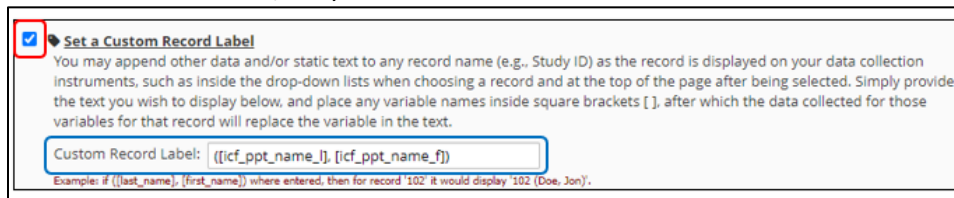


Figure 10. "Piping" participant name into Custom Record Label

7.3.2 Enable the File Version for “File Upload” fields?

Uncheck the box. This project uses File Upload fields to store the signed ICF and Attestation surveys. This project should not allow these to be removed.

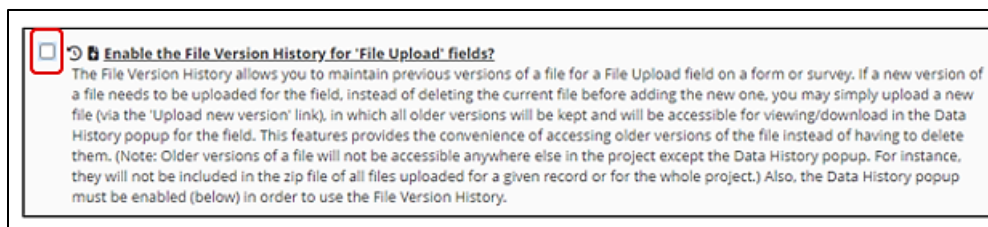


Figure 11. Uncheck "Enable the File Version History for "File Upload" fields"

7.3.3 Require a ‘reason’ when making changes to existing records?

Check the box. This project must require a reason if any existing record’s data is changed.

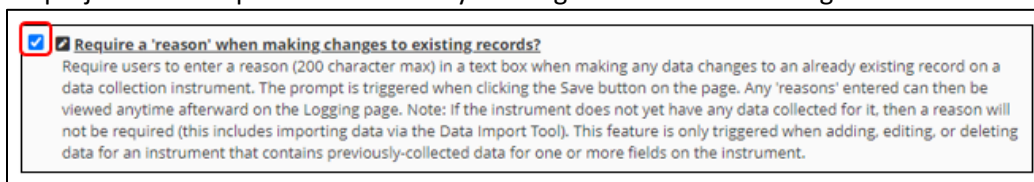


Figure 12. Check "Require a 'reason' when making changes to existing records"

7.4 Prepare IRB-approved consent PDF file for uploading to eConsent REDCap project

A copy of the IRB-approved ICF PDF file must be prepped so its individual pages can be uploaded to the ICF Survey instrument. *Note: if the IRB provides a “stamped” ICF PDF document, this will be the PDF file to use*

1) Save a copy of the approved ICF PDF file; name/save this file appropriately

a) **DO NOT alter the original document provided by IRB**

2) Open copy PDF in Adobe Acrobat Pro DC (*hereafter, “Acrobat”*)

3) Open the “Organize Pages” tool

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

- a) Allows extracting each page as an individual PDF file, automatically adding a “page #” identifier in the file name

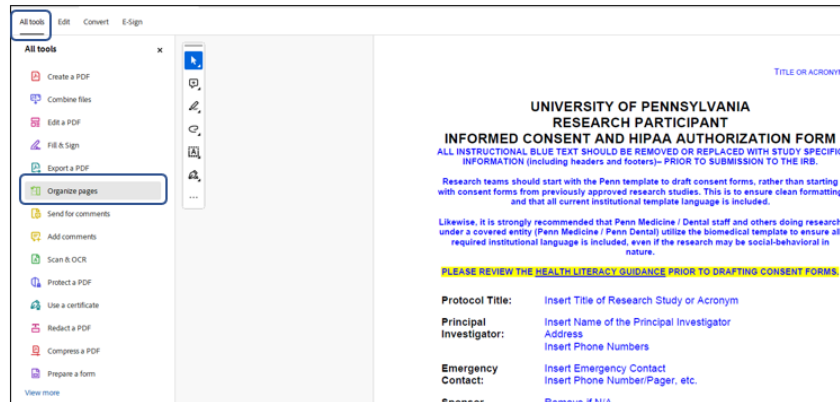


Figure 13. Navigation to Adobe's "Organize pages" tool

- b) Acrobat shows the below

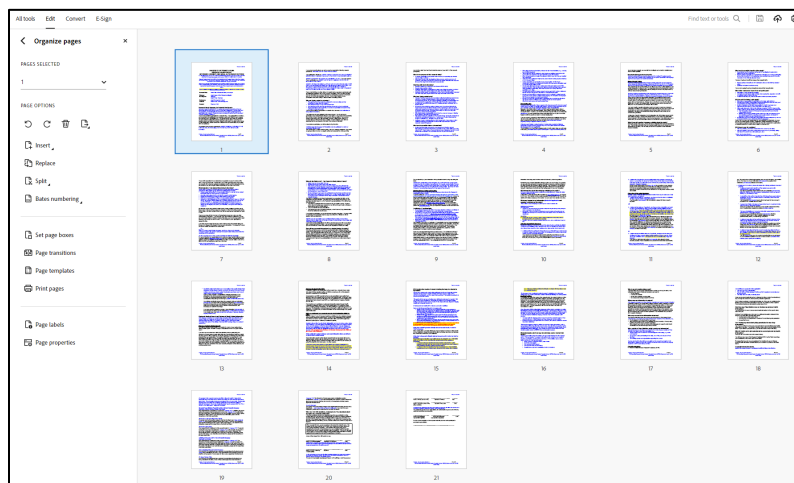


Figure 14. Adobe immediately after clicking "Organize pages"

- 4) Ensure page 1 is selected; click “Ctrl + A” to select all pages
5) Select “Extract” → Then check “Extract pages as separate files” → Click “Extract” button

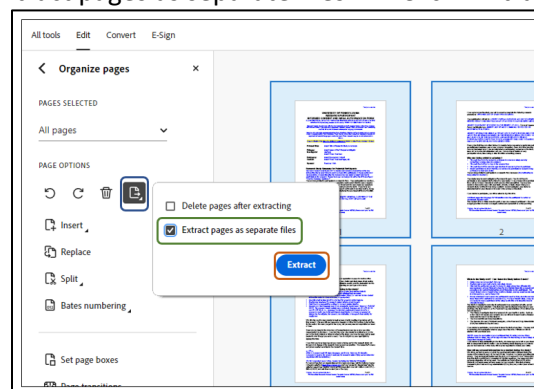


Figure 15. Navigation to extract each ICF page as an individual PDF file

- a) Save extracted files to same folder as copy of IRB-approved PDF; Acrobat saves all files with same file name but adds a space and a number to end of file name
6) Check folder and confirm that each page has its own PDF file

7.4.1 Convert each individual ICF page PDF file to image file

- 1) Open each ICF page PDF and convert them to an image file (see below screenshot)

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

a) Save each to an image file-specific sub-folder

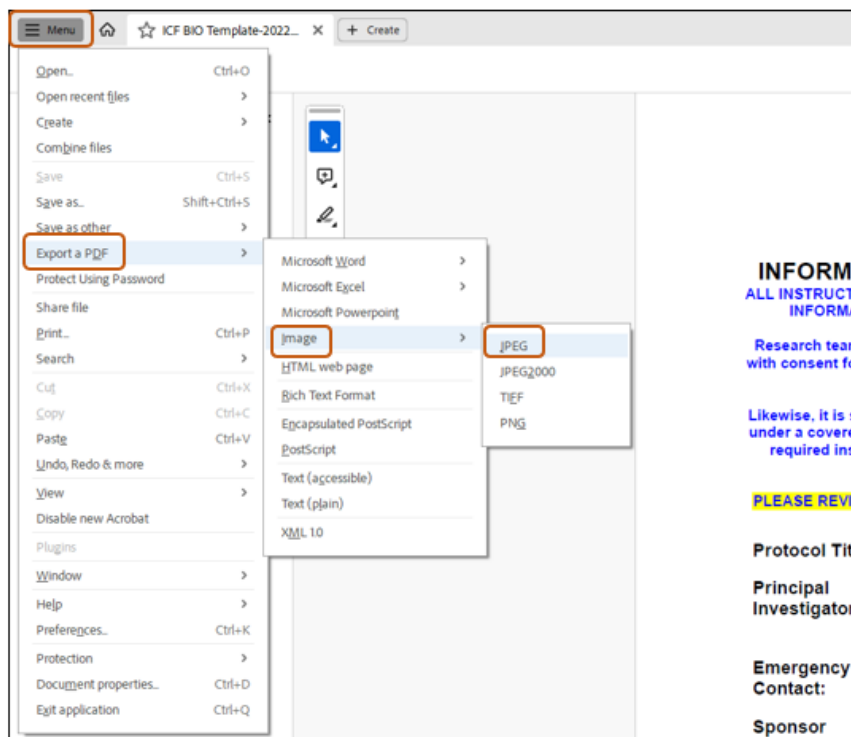


Figure 16. Navigation to export the PDF file as an JPEG/image file

2) Check folder and confirm that each page was converted to an image file

7.5 Create ICF_Survey instrument

7.5.1 Create “ICF_Survey” instrument & enable as survey

After “ICF_Survey” instrument creation, enable it as a survey by clicking the “Enable” button in the Online Designer. This will open the Survey Settings window and can be closed; Survey Settings are discussed later in this guidance.

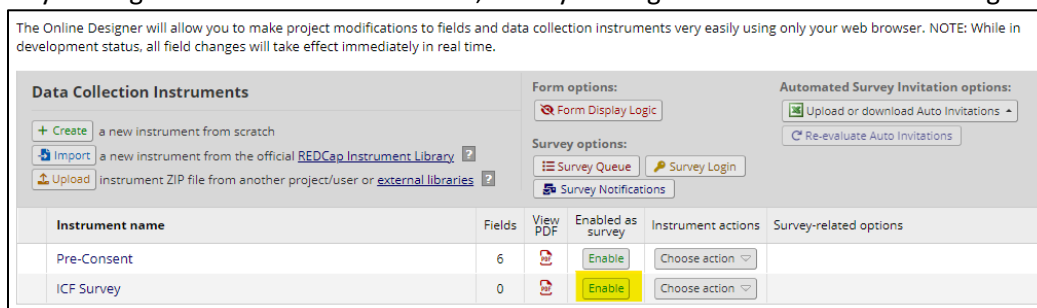


Figure 17. Click “Enable” button to make “ICF_Survey” into a survey

7.5.2 Create Descriptive Text fields and upload ICF image files

Uploading the ICF page image files ensures that the ICF pages are visible when participants view the ICF survey. REDCap’s Descriptive Text fields do accept uploaded PDF files; however, browser settings may block the visibility of PDF files.

1) Create a “Descriptive Text” field for each ICF page

a) Variable name = “**icf_v###_p###**” (“v###” = ICF version #; “_p###” = ICF’s page #; add leading zero for versions/pages 1 through 9]); Example: “icf_v01_p01”

i) Version # is needed to apply branching logic for future ICF versions

2) Steps to upload ICF page image files:

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

a) Click "Upload File"

Figure 18. Attach the ICF page image file

b) Click "Choose File" file and upload image file

Figure 19. Pop-up to attach ICF page image file

c) Select "Inline image/PDF" - this shows the uploaded ICF page on the survey

20. Orange box = attached file name; Green box = "Inline image/PDF" selection

d) Branching logic programming = "[icf_vers]='1'"

- i) Tells REDCap to only show this field if the record's [icf_vers] field's choice option equals Choice #1; this field is discussed [here](#)

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

- (1) The branching logic should be programmed so that the current ICF version, at the time of eConsent project creation, is selected (*i.e., if study's current ICF version is Version 3, branching logic should reflect that*)
- ii) ICF documents are often revised through the duration of the research study; therefore, it's best to set up all branching logic during initial project creation

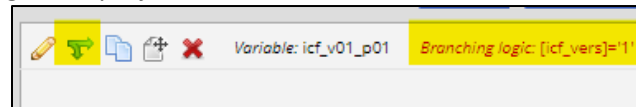
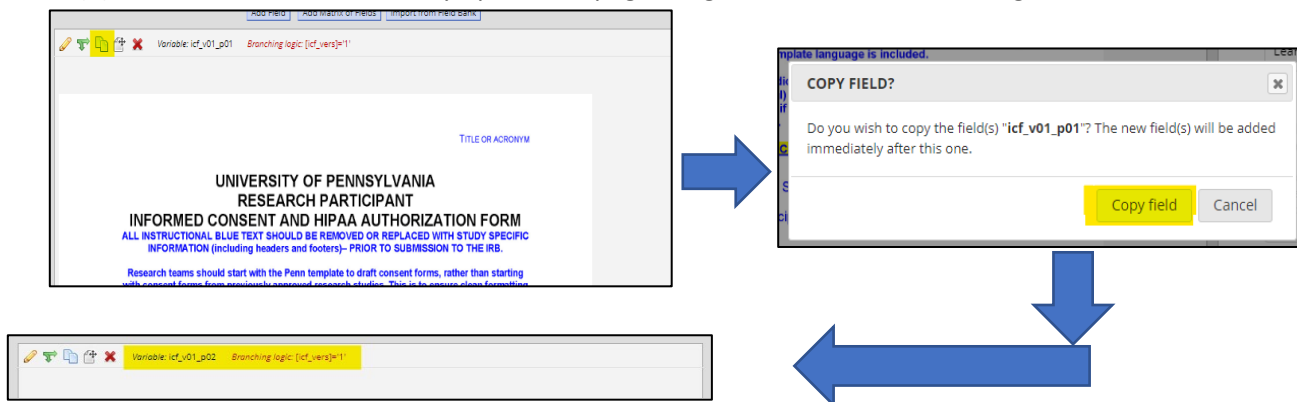


Figure 21. Branching logic programming

- e) Repeat process for remaining ICF pages
 - i) Click the **"Copy"** icon to create each remaining ICF page fields
 - (1) Copying automatically increases the variable name's **"_p##"** suffix by 1 number; also, the field's branching logic is copied and does not need to be fixed manually (*e.g., "icf_v01_p01" → click COPY → "icf_v01_p02"*)
 - (2) New/copied field appears immediately below the field from which it was copied
 - (3) Always copy the most recently created field
 - (4) Users still need to manually upload ICF page image file & set as "Inline image/PDF"



- 3) Create a "Begin New Section" field in before each Descriptive Text Field that contains a new page
 - a) Example: your project may have 3 Descriptive Text fields, each containing "page 1" of a different version of the ICF
 - i) DO NOT put the Begin New Section field between each "page 1" Descriptive Text field, put it before the first "page 2" Descriptive Text
 - b) Each **"icf_v##_p##"** field will be on its own page when viewed as a survey; branching logic ensures the correct version appears

7.5.3 Create remaining ICF_Survey fields

- 1) Descriptive Text field for the ICF's boxed **"When you sign this form..."** paragraph
 - a) Variable name: **"icf_box_msg"**
 - b) Check **"Use the Rich Text Editor"** → click the **"<>"** icon → copy/paste the below into the **"Source Code"** box

```
<div class="blue" style="text-align: center;">
<p>When you sign this form, you are agreeing to take part in this research study. s. Your signature also means that you are permitting the University of Pennsylvania to use your protected health information collected about you for research purposes within our institution. You are also allowing the University of Pennsylvania to disclose that protected health information to outside organizations or people involved with the operations of this study. </p>
<p>A copy of this consent form will be given to you.</p>
</div>
```

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

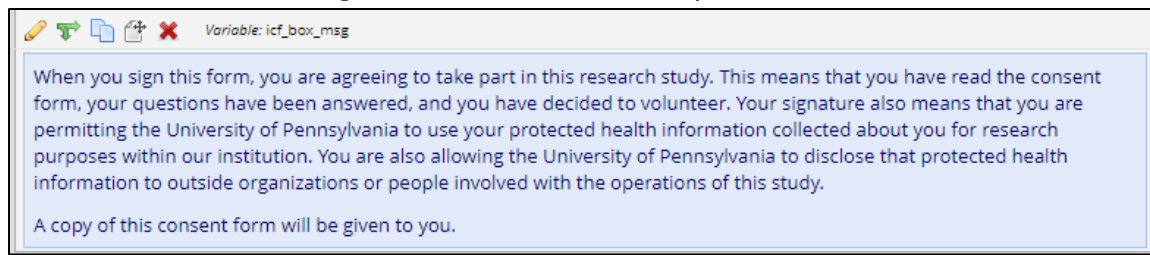


Figure 22. Descriptive Text field appearance using HTML code

- 2) Descriptive Text field to pipe participant's first and last name fields from the "Pre_Consent" instrument
 - a) Serves as the "Participant's Printed Name" section of ICF
 - b) Provides ICF version
 - c) Variable Name: "**pipe_ppt_name**"
 - d) Pipe the below fields into the Field Label; pipe fields by entering their variable names inside square brackets ("[]")

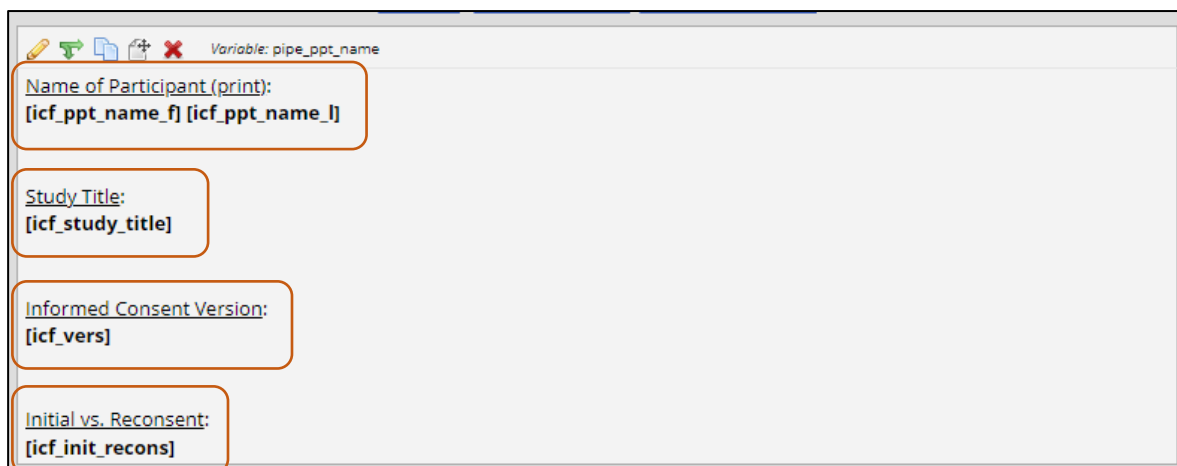
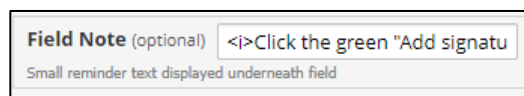
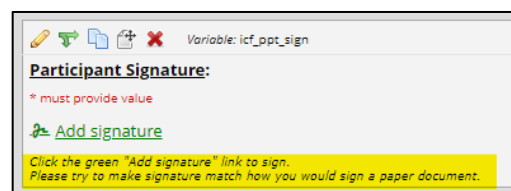


Figure 23. Spacing recommended to enhance appearance in PDF copies

- 3) Signature field to capture participant signature
 - a) Variable name: "**ifc_ppt_sign**"
 - b) Required = Yes; Identifier = Yes
 - c) Field Note: signatures should resemble natural signatures as much as possible and not be scribbled (*Copy/paste below HTML code*)

<i>Click the green "Add signature" link to sign.
Please try to make signature match how you would sign a paper document.</i>





- 4) Text Box field to capture date of participant's signature
 - a) Variable name: "**ifc_ppt_sign_date**"
 - b) Minimum validation: "[icf_prec_comp_dte]" – the earliest date entered must be on/after the date the record's "Pre_Consent" data entry form was completed
 - c) Maximum validation: "**today**" (type without quotes; if a future date is entered, error message appears)
 - d) Required = Yes; Identifier = Yes

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

- e) Optional recommendations
 - i) Using **@FORCE-MINMAX** Action Tag forces entered date to be within the Min/Max validation

The screenshot shows the 'Variable Name' field set to 'icf_ppt_sign_date'. Below it, the 'Validation?' dropdown is set to 'Date (M-D-Y)'. The 'Minimum' field is set to '[icf_prec_comp_dte]' and the 'Maximum' field is set to 'today'. There are also checkboxes for 'Enable auto naming of variable based upon its Field Label?' and 'Field Embedding'.

Figure 24. "Participant Signature Date" field's variable name & validation

7.5.4 ICF_Survey survey settings

- 1) Survey Title: "Study Electronic Informed Consent"
 - a) Rename if desired
- 2) Survey Instructions
 - a) Pipe **[icf_study_title]**, **[icf_irb_num]**, & **[icf_pi]** (fields automatically appear on PDF copies)
 - b) If desired, provide additional instructions for participants; delete the row if not needed

The screenshot shows the 'Survey Instructions' configuration. The 'Survey Title' is 'Study Electronic Informed Consent'. The 'Survey Instructions' field contains the following text: 'Study Title: [icf_study_title]', 'IRB Number: [icf_irb_num] | Principal Investigator: [icf_pi]', and 'ADD INSTRUCTIONS FOR TEAMS HERE'.

Figure 25. Survey Instructions using piped fields

- 3) Ensure the below Survey Settings are set
 - a) "Pagination" – "Multiple pages"
 - i) Check "Display page numbers at top of survey page"

The screenshot shows the 'Pagination' settings. The 'Multiple pages (display one section per page)' dropdown is selected. The checkbox 'Display page numbers at top of survey page' is checked. The checkbox 'Hide the 'Previous Page' button (i.e., Back button) on the survey page' is unchecked.

Figure 26. Multiple page survey settings

- b) "e-Consent Framework and PDF Auto-Archiver" – choose **"Auto-Archiver + e-Consent Framework"**
 - i) **Yellow highlighting = required settings**
 - (1) Setting the e-Consent Framework enables the end-of-survey certification and archival of PDF consent form
 - (2) "Allow e-Consent responses to be edited by users?" = UNCHECKED
 - (3) **First and Last name** fields: show the participant's name on the PDF consent forms
 - (4) **Signature field #1**: set to delete the participant's signature if they hit the "Previous Page" button once they reach the certification page
 - ii) **Green highlighting = optional**
 - (1) "e-Consent version" – shows the Consent version on the archived PDF files
 - (2) "e-Consent type" – may be useful if study uses different consents (i.e., separate consents used for specific study arms)

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

e-Consent Framework
- and -
PDF Auto-Archiver
Upon survey completion, a compact PDF copy of the survey response will be automatically stored in the project's File Repository, from which the archived PDFs can be downloaded at any time.

☐ Disabled
☐ Auto-Archiver enabled
☒ **Auto-Archiver + e-Consent Framework** [What is the e-Consent Framework?](#)
(includes end-of-survey certification & archival of PDF consent form)

e-Consent Framework Options:
For e-Consent it is sometimes required to include the consenting participant's name (and date of birth in some cases) on the final consent form as extra documentation of their identity. Below you may select fields used to capture that info. You may also enter the current e-Consent version and e-Consent type for this form. The values for the fields below will be automatically inserted into the footer of the PDF consent form that the participant will review at the end of the survey, after which that PDF 'hard-copy' will be archived in the File Repository. [Read more](#)

☐ Allow e-Consent responses to be edited by users?

e-Consent version:

First name field:

Last name field:

Note: If you are using a single field to capture whole name, you may select it for either first/last name above while leaving the other name field unselected.

Optional fields (these are not always necessary for e-Consent):

e-Consent type:

Date of birth field:

Force signature field(s) to be erased if participant clicks Previous Page button while on the certification page?
Select a field below that serves as a signature field in this survey. It could be a free-form text field, a signature field, or a number field (e.g., to collect a PIN), and it must be a Required field. If any fields are selected below, then if the participant gets to the last page of the survey where it asks them to certify their responses, if they then choose to click the Previous Page button, it will erase the value of these signature fields, thus forcing them to 'sign' the field(s) again before completing the survey. If you do not want this behavior, do not select any fields below. You may use up to five signature fields.

Signature field #1:

[+ Select another signature field](#)

Figure 27. eConsent Framework settings; Yellow = required; Green = optional

- 4) Optional settings: settings that can be set up
 - a) Question Numbering: choose **"Custom numbered"**
 - b) Allow **"Save & Return Later"** option: may be useful for occasional situations where consenting process cannot be completed during a visit
 - i) If setting this, leave sub-options unchecked
 - c) **Survey Completion Text**: provides instructions to assist participant with steps after survey completion
 - d) Logo: upload "Penn Medicine" logo; site - <https://uphsxnet.uphs.upenn.edu/marketing/branding/logos/>
- 5) Settings to NOT use: any setting not mentioned above are not needed or recommended

7.6 Create Attest_Survey instrument

The "Attest_Survey" instrument is used to capture the signature and signature date of the study team representative who is obtaining the participant's consent to enroll in the study. It captures participant questions, the answers provided to those questions, study representative's first & last name, signature, and the date they signed.

7.6.1 Create "Attest_Survey" instrument & enable as survey

After "Attest_Survey" instrument creation, enable it as a survey by clicking the "Enable" button in the Online Designer.

7.6.2 Create Attest_Survey fields

- 1) Descriptive Text field to remind study team to complete instrument as survey when they open "Attest_Survey" as a data entry form
 - a) Variable Name: **"warn_attest_open"**
 - b) Field Label: copy/paste the below into the **"Source Code"** box

```
<div class="red" style="text-align: center;">
<h2 style="margin: 0in;"><strong>Remember to open as a <span style="text-decoration: underline;">survey</span></strong></h2>
</div>
```

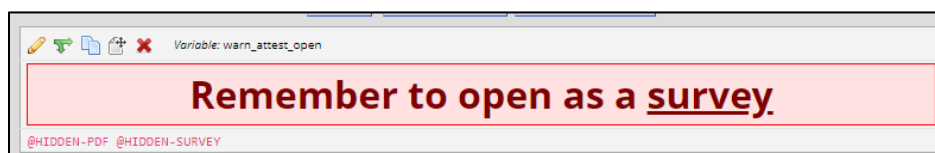


Figure 28. Descriptive Text field appearance using HTML code

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

- c) Action Tags: **@HIDDEN-PDF @HIDDEN-SURVEY**
- 2) Descriptive Text field to pipe PPT's name, date PPT signed ICF survey, Study Title, ICF Version, Initial vs. Reconsent, and the process by which informed consent was obtained (*i.e., Remote or In-person*)
 - a) Variable name: "**pipe_ppt_info_signed**"
 - b) Field Label text: pipe fields shown in below screenshot
 - c) Optional recommendations
 - i) Use Rich Text Editor to format Field Label text

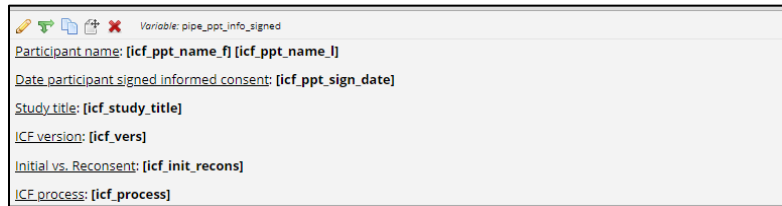


Figure 29. Descriptive Text field with piped fields; do this so fields appear on PDF copies

- 3) Yes/No Field: documents whether PPT had questions
 - a) Source field for branching logic to hide/unhide Notes Box field to capture questions and answers
 - b) Variable name: "**icf_ppt_ques**"
 - c) Required = Yes
 - d) Field Label: "Did the participant have any questions?"
- 4) Notes Box field to capture PPT's questions and provided answers
 - a) Variable name: "**icf_ppt_q_and_a**"
 - b) Required = Yes
 - c) Field Label: "Document all questions and the provided answers here:"
 - d) Branching Logic: **[icf_ppt_ques]='1'**

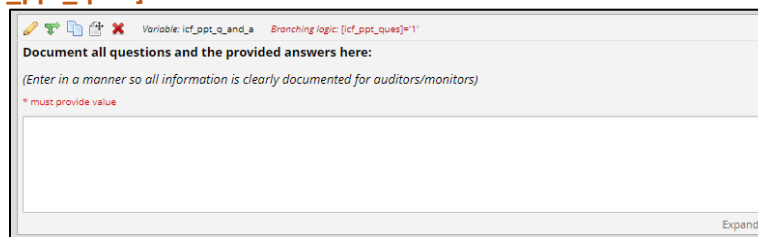


Figure 30. Notes Box to capture participant questions & answers

- 5) Descriptive Text field providing Study Rep Attestation statement
 - a) Variable name: "**attest_statement**"
 - b) Field Label: copy/paste the below into the "Source Code" box

```
<div class="blue" style="text-align: center;">
<h6 style="margin: 0in; text-align: left;"><strong>By signing below, I attest that:</strong></h6>
<ul>
<li style="text-align: left;"><strong>I have personally explained the research study to the above-named participant and answered all questions.
</strong></li>
<li style="text-align: left;"><strong>I believe that the participant understands what is involved in the study and freely agrees to
participate.</strong></li>
</ul>
</div>
```

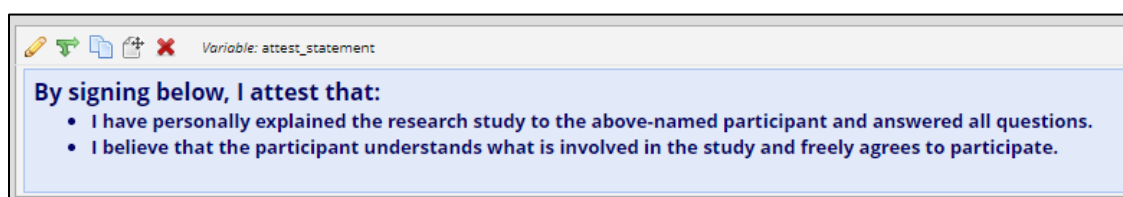


Figure 31. Descriptive Text field appearance using HTML code

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

6) Remaining fields

Variable Name	Field Type	Field Label Text	Validation	Action Tags	Notes
icf_rep_name_f	Text Box	Study Rep First Name	Required = Yes		Not PHI/PPI - Identifier tag not needed
icf_rep_name_l	Text Box	Study Rep Last Name	Required = Yes		Not PHI/PPI - Identifier tag not needed
icf_rep_sign	Signature	Study Representative Signature	Required = Yes		Field note: copy/paste below <i>Click the green "Add signature" link to sign. Please try to make signature match how you would sign a paper document.</i>
icf_rep_sign_date	Text Box	Study Representative Signature Date	Date (M-D-Y) Minimum: [icf_ppt_sign_date] Maximum: today Required = Yes	@FORCE-MINMAX	@FORCE-MINMAX forces date to be no earlier than PPT's signature date and no later than current date

Additional REDCap functionality can be used to enhance the look and flow of the form (*i.e.*, *Descriptive Text fields to provide instructions and/or warnings, Field Embedding, etc.*) Template project embeds Study Rep first and last name.

7.6.3 Attest_Survey survey settings

- 1) Survey Title: "Study Representative Attestation"
- 2) Survey Instructions
 - a) Pipe **[icf_study_title]**, **[icf_irb_num]**, and **[icf_pi]** (*fields automatically appear on PDF copies*)
 - b) Provide additional instructions for teams

Figure 32. Survey instructions with piped fields

- 3) Ensure below survey settings are set
 - a) "e-Consent Framework and PDF Auto-Archiver" = same as described for **"ICF Survey"**
- 4) Optional settings: = same as described for **"ICF Survey"**
- 5) Any setting not mentioned above are not needed or recommended

7.7 Create PDF_Uploads instrument

"PDF_Uploads" is a data entry form that captures PDFs of the signed eConsent and Study Attestation surveys in separate File Upload fields. REDCap automatically inserts each survey PDF after eConsent certification is completed. This form also contains date fields for tracking when PDFs of signed ICF documents are made available as well as downloaded by participants using REDCap's "Send-It" functionality. See [Section 8.2](#) for how to use Send-It.

While the eConsent Framework stores all completed PDFs in the project's File Repository, creating this instrument is highly recommended because it provides:

- Easy access to a specific participant's ICF documentation by connecting it to their project record (*Record ID*)
- Participant-specific workflow for tracking access to and download of the signed ICF PDF documents

7.7.1 Create PDF_Uploads fields

- 1) Create a Descriptive Text that provides instructions for using the "PDF_Uploads" form
 - a) Variable name: **"dtx_use_instr_pdf"**

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

b) Field Label: copy/paste the below into the "Source Code" box

```
<div class="red" style="text-align: center;">
<h6 style="text-align: center;"><span style="text-decoration: underline;"><strong>Use Instructions</strong></span></h6>
</div>
<hr>
<p><span style="font-weight: normal;">This instrument captures PDFs of the signed/certified "ICF_Consent" and "Attest_Survey" instruments,
providing easier access to the PDFs than searching in the PDF Survey Archive in the File Repository. </span></p>
<p><span style="font-weight: normal;">Access to the participant copies of these PDFs, via REDCap's "Send It" function, can be set up directly from
here. For each attached PDF, click the "Send It" link. Send It function opens in a new window. Follow OCR guidance for more information on using
Send It.</span></p>
<p><span style="font-weight: normal;">The date fields below provide a workflow for tracking access and receipt of the signed PDFs:</span></p>
<ol>
<li><span style="font-weight: normal;">The date PDF access was provided using Send It (for each PDF)</span></li>
<li><span style="font-weight: normal;">The date each PDF was downloaded by the recipient</span></li>
</ol>
<p><span style="font-weight: normal;">Once all fields are completed, change the form's "Complete?" status to "Complete." This serves as
confirmation that further action for the record is not required. </span></p>
```

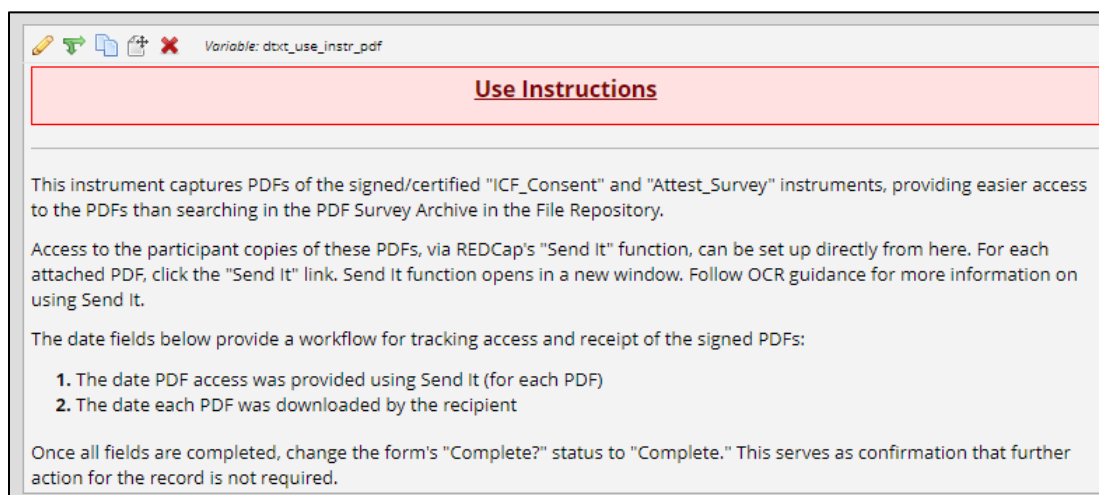


Figure 33. "Use Instructions" descriptive text field

2) Create remaining fields per the table below:

Variable Name	Field Type	Field Label Text	Validation	Action Tags	Notes
N/A	Begin New Section	Signed PDF files	N/A	N/A	Groups File Upload fields together on the form
icf_upload	File Upload	Uploaded signed eConsent PDF:	Required = Yes Identifier = Yes	None	Field Note: "Attached by REDCap automatically."
attest_upload	File Upload	Uploaded signed Attestation PDF:	Required = Yes Identifier = Yes	None	Field Note: "Attached by REDCap automatically."
N/A	Begin New Section	PDF access via Send-It	N/A	N/A	Groups the 2 date fields together on the form
sendit_dte_signed_icf	Text Box	Enter date that signed eConsent access set up via Send-It:	Date (M-D-Y) Minimum = [icf_rep_sign_date] Maximum = today	None	
sendit_dte_signed_attest	Text Box	Enter date that signed Attestation access set up via Send-It:	Date (M-D-Y) Minimum = [icf_rep_sign_date] Maximum = today	None	
N/A	Begin New Section	Signed PDF files downloaded	N/A	N/A	Groups the 2 date fields together on the form
dwnld_dte_signed_icf	Text Box	Date PPT downloaded eConsent from Send-It:	Date (M-D-Y) Minimum = [sendit_dte_signed_icf] Maximum = today	None	Field Note: "Date provided via Send-It email."

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

Variable Name	Field Type	Field Label Text	Validation	Action Tags	Notes
dwnld_email_conf_icf	Upload File	Upload a PDF of the Send-It confirmation email that the eConsent was downloaded:	N/A	None	
dwnld_dte_signed_attest	Text Box	Date PPT downloaded Attestation from Send-It:	Date (M-D-Y) Minimum = [sendit_dte_signed_attest] Maximum = today	None	Field Note: "Date provided via Send-It email."
dwnld_email_conf_attest	Upload File	Upload a PDF of the Send-It confirmation email that the Attestation was downloaded:	N/A	None	

7.7.2 Update "ICF_Survey" & "Attest_Survey" Survey Settings for upload to PDF_Uploads

- 1) "ICF_Survey" Survey Settings: "Survey Customizations" section → "Save a PDF of completed survey response to a File Upload field" dropdown → select **"icf_upload "Signed eConsent upload"**

Survey Customizations:

Question Numbering
For custom numbering, each question's number will be blank by default until manually added via the Online Designer or Data Dictionary.
Custom numbered ▼

Pagination
One page or multiple pages? Section headers, which begin new sections on the instrument, will serve as the page break in a multi-page survey, in which each page will begin with a section header.
Multiple pages (display one section per page) ▼
☒ Display page numbers at top of survey page
☐ Hide the 'Previous Page' button (i.e., Back button) on the survey page (prevents respondents from going back to previous pages)

Allow participants to download a PDF of their responses at end of survey?
Display a button for the participant to download a PDF file of their responses for the survey they just completed.
No ▼
This option will not be available if the Survey Auto-continue or Survey Queue auto-start option is enabled. Also, if a field utilizes the @HIDDEN action tag, it will not be displayed in the PDF.
Note: Because the e-Consent Framework option is enabled on this page, the PDF included here will not be the full-length PDF but will be the 'compact' PDF, which omits unanswered questions and unselected choices.

Save a PDF of completed survey response to a File Upload field
A PDF copy of the survey response will be immediately stored in the field selected on the right whenever a participant completes this survey.
icf_upload "Signed eConsent upload" ▼
Disabled
PDF_Uploads
icf_upload "Signed eConsent upload"
attest_upload "Signed study attestation upload"

Figure 34. Saving completed eConsent survey PDF to appropriate File Upload field

- 2) "Attest_Survey" Survey Settings

Save a PDF of completed survey response to a File Upload field
A PDF copy of the survey response will be immediately stored in the field selected on the right whenever a participant completes this survey.
attest_upload "Signed study attestation upload" ▼
Disabled
PDF_Uploads
icf_upload "Signed eConsent upload"
attest_upload "Signed study attestation upload"

Figure 35. Saving completed Attestation survey PDF to appropriate File Upload field

7.8 Set up "Alerts & Notifications"

Create the below alert in "Alerts & Notifications" to send PPTs survey invites for their "ICF_Survey" instrument. When the "Pre_Concent" form is completed, REDCap emails a survey invitation to PPTs using the email captured in the **"icf_ppt_email"** field.

- 1) Click "Alerts & Notifications" under "Applications" in the left-hand Project Menu
- 2) Click **"Add New Alert"** button
- 3) Enter informative alert title (i.e., **"SEND ICF SURVEY LINK TO PPT"**)
- 4) Step 1 settings
 - a) Alert triggers when the "Pre_Concent" instrument is saved with a COMPLETE status and the 2 email fields are identical; checking the "Ensure logic" box makes REDCap review the logic before sending the email

Figure 36. Step 1 settings highlighted

- a) These options can be changed to better suit study team workflows

Figure 37. Step 2 settings highlighted

- a) "Email From" = name and email that is sending the email to the PPT
- b) "Email To" = the PPT's email field entered in "Pre_Consent" instrument
- c) "Subject" = pipes in the study's title
- d) Check the **"Prevent piping of data..."** box helps protect any IDENTIFIERS piped into the email's body
- e) "Message" = enter your own email body
 - i) Include the **"[survey-link:icf_survey]"** Smart Variable which provides the PPT a unique survey link to the "ICF Survey" instrument connected to their project record

Figure 38. Step 3 settings highlighted

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

Users in the “Full Rights” User Role have the highest-level privileges, such as the ability to delete records, add/remove project users, and revise the project’s fields; therefore, this Role should be restricted to only one or two study team members.

Users in the “Attestation” User Role only have the permissions needed to create new project records and complete the “PDF_Uploads” form.

Screenshots for both User Roles can be found in the [Appendices](#).

7.10 Testing and move to Production status

Thoroughly test your project using mock data to make sure everything works. **“Everything”** includes:

- All fields capture what they are supposed to capture (*e.g., date fields capture only dates, Upload File fields capture signed PDFs*)
- All field validation works as expected (*e.g., MIN/MAX validation*)
- Branching logic
- Alert invitation email to PPTs
- eConsent Framework is set up correctly (*e.g., certification step appears/works, PDFs appear in File Repository*)
- Send-It works as expected

Testing may identify additional project flaws that require resolution. Post-testing revisions may require additional testing to ensure proper functioning.

Once everything is THOROUGHLY tested, including post-testing revisions/testing, request the project be moved to Production status.

Ask study team members to test/provide feedback (*e.g., enhancements, things that don’t work properly, etc.*).

NEVER ENTER REAL PPT eConsent data when the project is in Development status.

8 eConsent Workflows

There are multiple ways in which study teams can conduct the ICF process electronically. This eConsent Template REDCap project can serve most eConsent workflows. For example, study teams could complete the “Pre_Consent” instrument prior to the study visit during which the ICF process is completed. This would provide participants with a copy of the ICF to review; however, it increases the chances that a participant may forget to ask any questions during the ICF process. Further, this workflow can also lead to participants e-signing the ICF before it’s thoroughly reviewed by the study team.

OCR Operations recommends that the eConsent process mirror an in-person consenting process as much as possible. Sending copies of the ICF to participants ahead of study visits is not a problem; however, they shouldn’t be a copy that can be signed by participants. OCR Operations recommendations are listed below.

8.1 Recommended eConsent Workflows using this REDCap template

- 1) Follow Penn Medicine remote consent guidance; follow all federal, state, and local laws/regulations
 - a) Check UPenn IRB and OCR websites for updated guidance
 - b) Remote consents should be completed at a scheduled visit, just as you would with an in-person consent, using an appropriate platform
- 2) Do not complete the “Pre_Consent” instrument until PPT has answered call/logged on to video conferencing platform/etc.
 - a) Prevents creation of project records if PPT misses visit

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

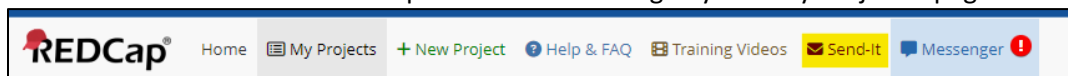
- 3) Confirm the person on the call/video conference is the actual participant
 - a) Ask PPT to provide official identification (*e.g., state-issued driver's license*) by showing it on the video camera
- 4) Capture the correct email address in "Pre_Consent"
- 5) Confirm that the participant has received the "ICF_Survey" survey on the call/video conference
- 6) Reminders
 - a) Zoom: DO NOT record; recording is NOT HIPAA-compliant
 - b) Microsoft Teams: preferred solution for UPHS employees
 - c) Be sure to follow all guidance regarding recordings
- 7) Study teams should remain on the call to answer any PPT questions
- 8) Provide "Survey Completion" instructions to let PPTs know how to inform study teams they have finished the "ICF_Survey"
 - a) Study teams can determine this by refreshing the eConsent REDCap project's Record Home Page/Record Status Dashboard
- 9) Study teams complete the "Attest_Survey" survey as soon as PPTs complete their "ICF_Survey," while the PPT is still on the call
 - a) Allows study team to document all questions and answers
- 10) Create standard processes regarding long-term storage and providing PPTs copies of the signed ICF/Attestation Survey PDFs; include:
 - a) Where PDFs will be stored
 - i) Penn Medicine REDCap is HIPAA-compliant and is an appropriate storage location
 - ii) If the primary storage location is NOT Penn Medicine REDCap, the location must be compliant with HIPAA
 - b) How PDFs will be shared with PPTs
 - i) See [Section 8.2](#) for how to use REDCap's Send-It function
 - ii) Emailing them directly to PPTs **is not** a secure sending method
 - c) Study-specific procedures should document these standard processes and stored with other study-related procedures

8.2 REDCap's Send-It

Send-It is a secure data transfer application that allows you to upload a file (up to 512 MB in size) and then allow multiple recipients to download the file in a secure manner. Each recipient will receive:

- 1) An email containing a unique download link/URL
- 2) A second follow-up email with the password (for greater security) for downloading the file

The file will be stored securely and then later removed from the server after the specified expiration date. For more information, Send-It function is located on the top menu when looking at your "My Projects" page in REDCap:



8.2.1 Using Send-It: providing signed informed consent PDFs

Survey PDFs can be sent via Send-It from a PPT's PDF_Uploads form using their Record Home Page.

- 1) Navigate to PPT's Record Home Page and click the green bubble next to "PDF_Uploads"

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

Record Home Page

The grid below displays the form-by-form progress of data entered for the currently selected record. You may click on the colored status icons to access that form/event.

[Choose action for record](#)

Record ID 1 (Sanborn-01, Paul-Test)

Data Collection Instrument	Status
Pre_Consent	
ICF_Survey (survey)	
Attest_Survey (survey)	
PDF_Uploads	

Legend for status icons:

- Incomplete
- Incomplete (no data saved) ?
- Unverified
- Partial Survey Response
- Complete
- Completed Survey Response

Figure 39. Navigation to a record's "PDF_Uploads" instrument

- 2) Click the "Send-It" link under a PDF link

Signed PDF files

Uploaded signed eConsent PDF:

[StudyElectronicInformedCo... 0906.pdf \(7.96 MB\)](#)

[Remove file](#) or [Send-It](#)

Project uploads PDF automatically after participant submits the ICF Survey.

Uploaded signed Attestation PDF:

[StudyRepresentativeAttest... 0908.pdf \(0.04 MB\)](#)

[Remove file](#) or [Send-It](#)

Project uploads PDF automatically after study team representative submits the Attestation Survey.

Figure 40. Click "Send-It" link to start process.

- 3) Opens Send-It pop-up window
 - a) Select appropriate "From:" email address
 - i) Lists the email addresses associated with your REDCap User account
 - (1) Sender's email address must end in either "@pennmedicine.upenn.edu" or "@upenn.edu"
 - (2) Gmail and other email services may block emails coming from other email addresses
 - b) "To:" = Enter the PPT's email
 - i) Cannot select the PPT's email address captured in eConsent project
 - c) Provide an optional "Email subject" & "Email message"
 - d) Select appropriate "Expiration" (range = 1 to 14 days)
 - e) Check the "Receive confirmation?" box – REDCap will email a notification to the sender when the recipient downloads the document
 - f) Click "Send It!" button

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

REDCap - Google Chrome

https://redcap.med.upenn.edu/redcap_v13.0.2/index.php?route=SendItController:upload&loc=3&id=1488830&spid=32329

REDCap® Home My Projects + New Project Help & FAQ Training Videos Messenger Logged in as sanborn More

✉ Send-It: Sending file "PUTSTUDYTITHEREEElectronicInf_2022-09-09_1555.pdf"

Instructions for using Send-It:
Fill out the form below to specify to whom you wish to send the file, as well as other custom information and settings. Each email address you enter below will receive a message stating that a file is available for download. The email will include a unique link and password, which the recipient will use to navigate to the webpage for downloading the file to their computer. If the file is very large, it may take several moments to upload, so please allow it to continue to upload until it notifies you of its completion.

From: sanborn@upenn.edu

To: sanborn@penmedicine.upenn.edu
(recipient emails)
Separate email addresses with commas, semi-colons, or line breaks

Email subject: Here's the PDF copy of your signed consent
(optional)

Email message: Hi,
Attached is your copy of the signed informed consent document for our research study.
Thanks!
(optional)

Expiration: 3 days Specify the time after which the file will no longer be accessible for download

File: PUTSTUDYTITHEREEElectronicInf_2022-09-09_1555.pdf (22.93 MB)

☒ **Receive confirmation?**
Get an email notification informing you when your file has been downloaded by each recipient.

Send It!

REDCap 13.0.2 - © 2022 Vanderbilt University

Figure 41. Send-It pop-up menu settings highlighted

4) Pop-up changes with confirmation that the PDF was successfully uploaded

REDCap - Google Chrome

https://redcap.med.upenn.edu/redcap_v13.0.2/index.php?route=SendItController:upload&loc=3&id=1488830&spid=32329

REDCap® Home My Projects + New Project Help & FAQ Training Videos Messenger Logged in as sanborn More

✉ Send-It: Sending file "PUTSTUDYTITHEREEElectronicInf_2022-09-09_1555.pdf"

✔ **File successfully uploaded!**

The file "PUTSTUDYTITHEREEElectronicInf_2022-09-09_1555.pdf" (22.93 MB) was successfully uploaded, and the recipients listed below were sent an email with a password and unique link for downloading the file. The file will expire and become inaccessible on Sunday, 12/25/2022 9:38am.

Recipients:
sanborn@penmedicine.upenn.edu

REDCap 13.0.2 - © 2022 Vanderbilt University

Figure 42. Send-It confirmation screen

- 5) Close pop-up to go back to record's "PDF_Uploads" form
- 6) Enter the date access to the PDF files was provided to the participant

PDF access via Send-It

Enter date that signed eConsent access set up via Send-It:

12-08-2023 Today M-D-Y

Enter date that signed Attestation access set up via Send-It:

12-08-2023 Today M-D-Y

Figure 43. Enter dates that access to signed PDF files provided to participant

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

8.2.2 Using Send-It: participant receipt of signed PDF copies

1) For each PDF sent via Send-It, recipient receives 2 emails

a) One with the link to the PDF

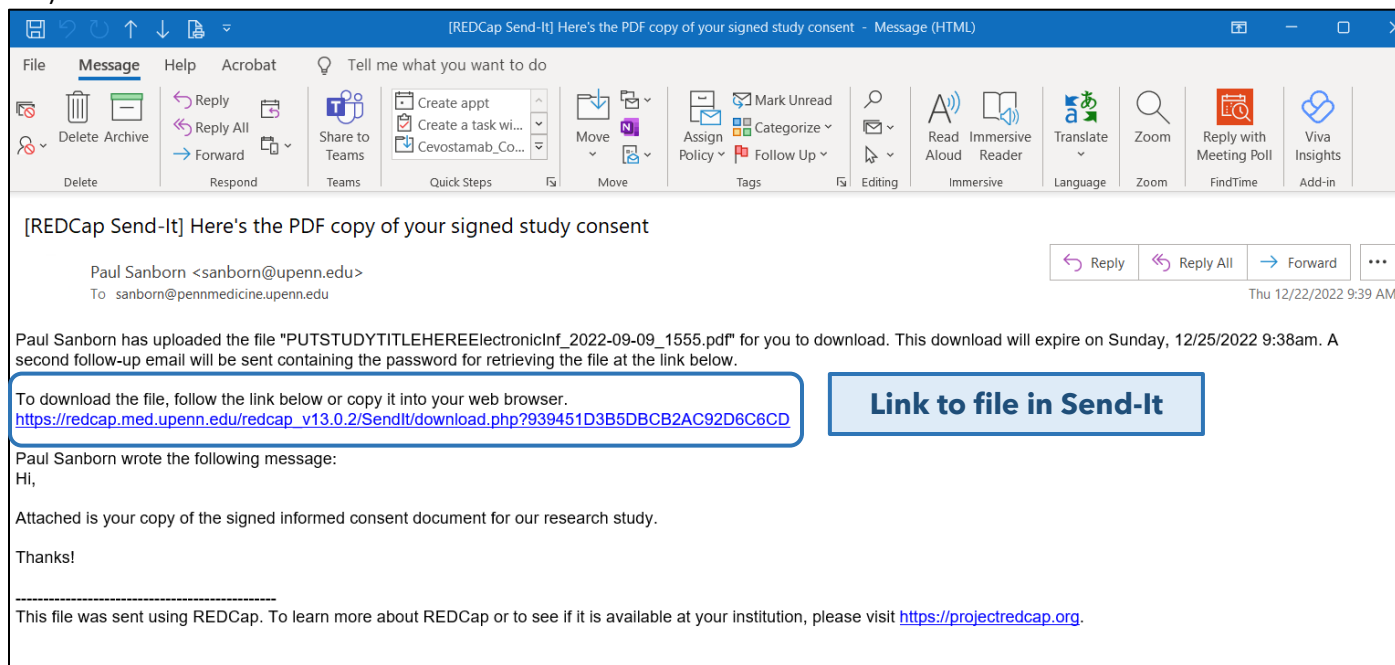


Figure 44. First recipient email contains link to Send-It file

b) Second with the password needed to download

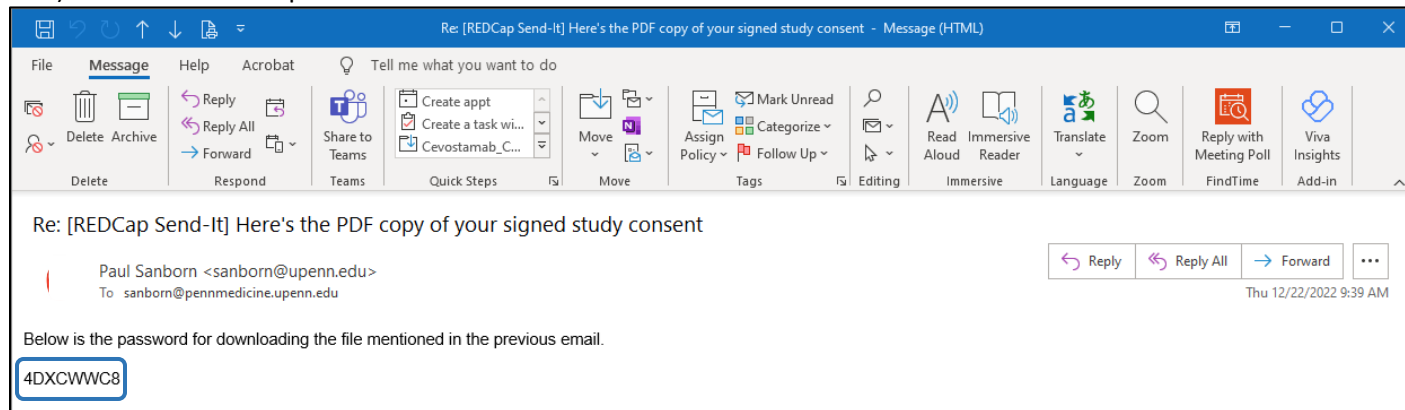
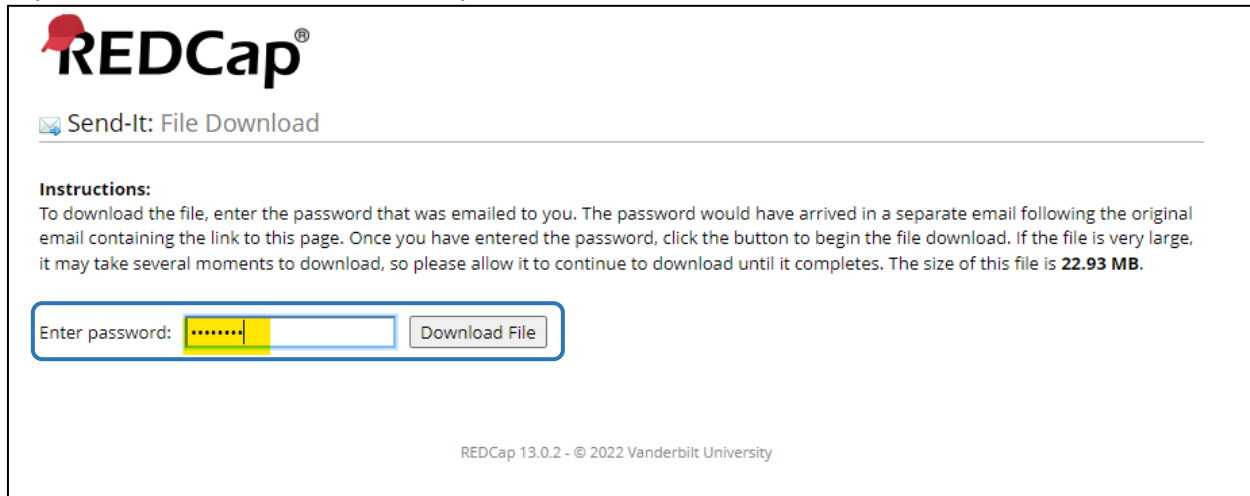


Figure 45. Second recipient email contains password to download file

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

- 2) Participant clicks link in first email, enters password, and clicks “Download File” button



REDCap®

Send-It: File Download

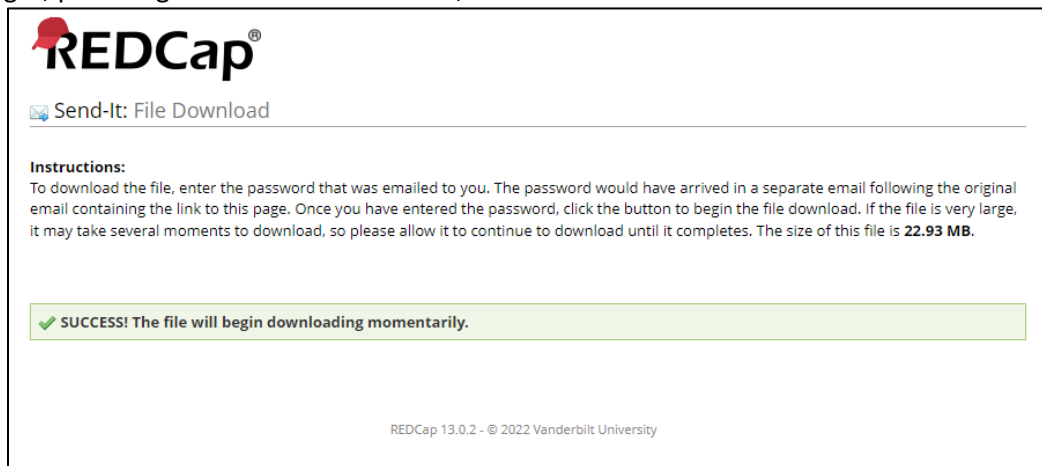
Instructions:
To download the file, enter the password that was emailed to you. The password would have arrived in a separate email following the original email containing the link to this page. Once you have entered the password, click the button to begin the file download. If the file is very large, it may take several moments to download, so please allow it to continue to download until it completes. The size of this file is **22.93 MB**.

Enter password:

REDCap 13.0.2 - © 2022 Vanderbilt University

Figure 46. PPT enters provided password & clicks Download File

- 3) Screen changes, providing download confirmation; PDF downloaded to device



REDCap®

Send-It: File Download

Instructions:
To download the file, enter the password that was emailed to you. The password would have arrived in a separate email following the original email containing the link to this page. Once you have entered the password, click the button to begin the file download. If the file is very large, it may take several moments to download, so please allow it to continue to download until it completes. The size of this file is **22.93 MB**.

✓ **SUCCESS! The file will begin downloading momentarily.**

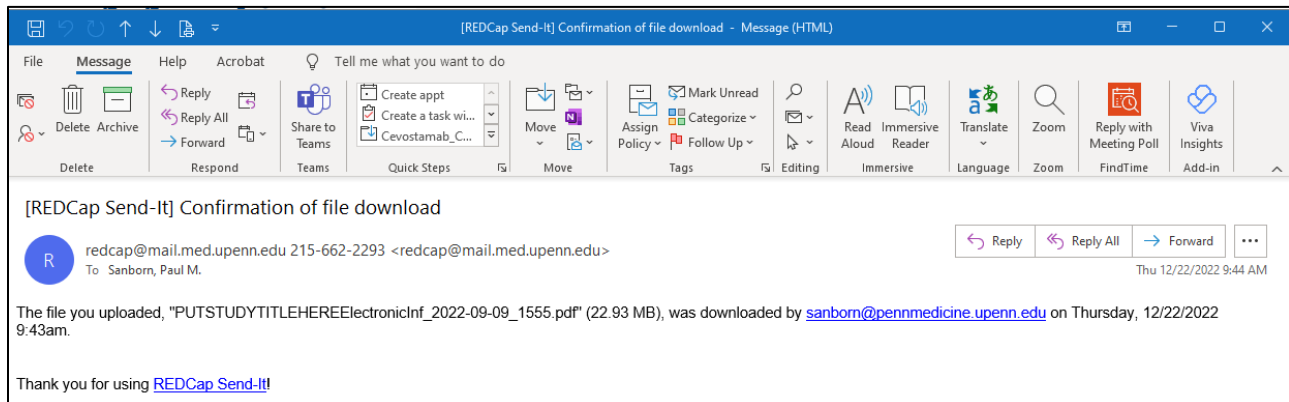
REDCap 13.0.2 - © 2022 Vanderbilt University

Figure 47. Screen confirming document downloaded to device

8.2.3 Using Send-It: confirmation that participant downloaded files

- 1) By checking the “**Receive Confirmation**” checkbox, REDCap will email the address provided described in [Section 8.2.1](#)

- a) Email in the “From:” field receives the below confirmation when file is **downloaded**



[REDCap Send-It] Confirmation of file download - Message (HTML)

File Message Help Acrobat Tell me what you want to do

Delete Archive Reply Reply All Forward Share to Teams Quick Steps Move Tags Editing Immersive Language Zoom FindTime Viva Insights Add-in

[REDCap Send-It] Confirmation of file download

redcap@mail.med.upenn.edu 215-662-2293 <redcap@mail.med.upenn.edu>
To: Sanborn, Paul M.

The file you uploaded, "PUTSTUDYTITLEHEREElectronicInf_2022-09-09_1555.pdf" (22.93 MB), was downloaded by sanborn@pennmedicine.upenn.edu on Thursday, 12/22/2022 9:43am.

Thank you for using [REDCap Send-It!](#)

Figure 48. Email confirmation sent to sender when file is downloaded

- 2) When these are received, study team must navigate to the record’s “PDF_Uploads” form and enter the dates the PDF files were downloaded

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

3) For additional confirmation, upload a PDF of the Send-It confirmation email

Figure 49. Enter dates for when participant downloaded the PDF files via Send-It

4) Repeat process for the Attestation PDFs

9 Modified informed consent versions

Including the “**icf_vers**” field in the “Pre_Consent” instrument when building REDCap eConsent projects allows study teams to include new, IRB-approved ICF versions with minimal project revisions.

9.1 Take project offline & communicate with study team

It’s best to take the project offline to make all project revisions. While project is offline, it cannot be used to complete any consents. For this reason, inform your study team when and how long the project will be offline.

9.2 Make backup copy of project

First step is to make a copy of the eConsent project, including all data. This provides a backup of the original project, in its pre-revision state, in case mistakes made during revisions are not identified before moving the project back to Production Status.

9.2.1 Steps

- 1) Login to Penn Medicine REDCap
- 2) Click eConsent project’s name on the “My Projects” page
- 3) Click “Other Functionality” tab
- 4) Click green “Copy the Project” button (“Copy or Back Up the Project” section)
 - a) Add “_COPY” to “Project title:”
 - b) “Also copy the following:” checkboxes
 - i) Check the “All records”
 - ii) Uncheck “Users and their user rights” checkbox; study team users can be added later, if needed
 - iii) Leave other checked options checked

Copied project will appear on your “My Projects” page. If anything goes wrong with revisions, use this copy to start over.

9.3 Make copy of the project to complete & test revisions

Revising projects that have been moved to Production status cannot be tested during the revisions because the revisions aren’t visible until the project moves back to Production status.

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

Make another full copy of the project to make and test all revisions. Once the revisions have been made and thoroughly tested, revise the real eConsent project.

While it seems inefficient, it's more efficient because there is no lag time while you wait for REDCap Administrators to move the project back to Production status to test the revisions. Inevitably, there is always something that doesn't work during testing in which you'll have to move the project back to Development to make the corrections, then request the move back to Production, wait, then test again.

Time is saved by making/testing revisions in a copy, then repeating the revisions in the "real" project.

9.3.1 Steps

Follow all steps in [Section 9.2.1](#) with the below exceptions:

- Add "_REVISIONS" to "Project title:"
- "Also copy the following:" checkboxes
 - Leave "All records" unchecked
 - Uncheck "Users and their user rights"
 - Leave other checked options checked

9.3.2 Preparing the revised IRB-approved modified ICF version for uploading

Same exact process as [Section 7.3](#). Be sure all file names indicate the ICF's modified version.

9.3.3 "Pre_Consent" instrument changes

Revise the "icf_vers" field.

1) Choice options

- a) Add choice option for new version
 - i) Choice code = Next available sequential number
 - ii) Choice label = "New version #" + "(" + "Start date" for new version + "-" + "End Date" + ")"

b) Always use the IRB-approval periods provided on the IRB-approved ICF document

- i) If IRB did not provide an approval end date, leave it blank in the choice option (*may apply to protocols not requiring Continuing Review*)
 - (1) When revised ICFs are approved, enter the end date in the previous version's Choice label

Figure 50. Choice options with both Start and End Dates

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

Figure 51. Choice options (Current version doesn't have End Date)

- 2) Action Tag: change the **@DEFAULT** Action Tag's logic to the new version's choice code (example: **@DEFAULT='3'**)

9.3.4 "ICF_Survey" instrument changes

Create Descriptive Text fields to contain the image file for each page of the new ICF version. There are several ways to do this:

- 1) Complete the same steps outlined in [Section 7.4.2](#).
 - a) Variable name = **"icf_v##_p##"** with **"_v##"** being the new ICF version number
- 2) Make a copy of the previous version's first page and alter for new version:
 - a) Change variable name to follow new version format outlined in #1 above
 - b) Remove the previous version's image file

- c) Upload the Page 1 image file of the new ICF version
 - d) Update field's branching logic to show only if **"[icf_vers]='#'"** ("#" being new version's choice code)
 - e) Then make a copy of this new field for each page of the revised ICF (as in [Section 7.4.2](#))
- 3) Update the "e-Consent version" field in the Survey Settings (if used to document consent versioning)

9.3.5 "Attest_Survey" instrument changes

No changes required in this instrument unless the Survey Setting's "e-Consent version" field is used to document consent versioning.

9.3.6 Test revisions in the copy project

Same testing process as outlined in [Section 7.8](#).

9.4 Revise the "live" eConsent project

Revisions in the "live" project cannot be made until after they have been thoroughly tested in the copy. This ensures that the revisions will not affect the data captured in the "live" eConsent project. Move the real project to Development mode, make all revisions, and request the move to Production status.

After the "live" eConsent project is in Production status, both copies can be deleted.

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

10 Reconsenting

When used as a project separate from study data, reconsenting participants can be completed using this eConsent REDCap project easily. These projects permit individual participants having multiple eConsent records capturing reconsents. Creating reports to track reconsented participants is easier because there are fewer fields and filtering required. This may not be the case if an eConsent project is incorporated into a REDCap project capturing study data.

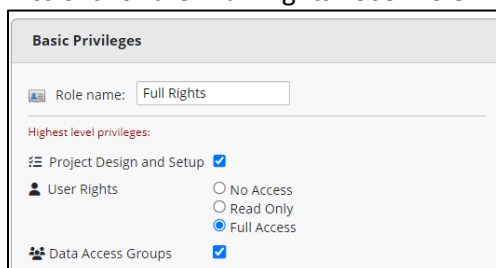
When eConsent tracking is incorporated into projects also capturing research study data, reconsenting can be more challenging. While using longitudinal data collection and/or repeated instruments/events to capture reconsents is possible, their complexity requires more effort (and time) to build and test. The “signature” settings in the eConsent Framework are specific to the project’s events; therefore, teams would need to build at least two sets of survey instruments for the “ICF_Survey” and the “Attest_Survey” – one for the initial consent and one for reconsents. The duplicated sets then require revisions to both sets for new ICF versions. Essentially, these types of projects are more complicated than needed for the purpose of capturing eConsents.

Reconsenting can be easily done by creating a new record and selecting the “Reconsent” option in the “**icf_init_recons**” field on the “Pre_Consent” instrument. The rest of the process is the same as if it was an initial consent.

11 Appendices

11.1 “Full Rights” User Role permissions

The below screenshots provide the permissions for the “Full Rights” User Role.



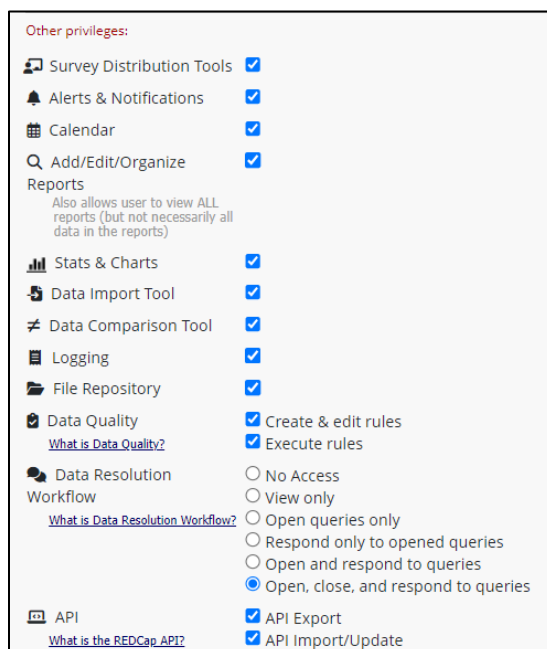
Basic Privileges

Role name:

Highest level privileges:

- Project Design and Setup ☒
- User Rights
 - ☐ No Access
 - ☐ Read Only
 - ☒ Full Access
- Data Access Groups ☒

Figure 52. “Full Rights” User Role | Name & Highest-level privileges



Other privileges:

- Survey Distribution Tools ☒
- Alerts & Notifications ☒
- Calendar ☒
- Add/Edit/Organize ☒
- Reports
 - Also allows user to view ALL reports (but not necessarily all data in the reports)
- Stats & Charts ☒
- Data Import Tool ☒
- Data Comparison Tool ☒
- Logging ☒
- File Repository ☒
- Data Quality
 - [What is Data Quality?](#)
 - ☒ Create & edit rules
 - ☒ Execute rules
- Data Resolution Workflow
 - [What is Data Resolution Workflow?](#)
 - ☐ No Access
 - ☐ View only
 - ☐ Open queries only
 - ☐ Respond only to opened queries
 - ☐ Open and respond to queries
 - ☒ Open, close, and respond to queries
- API
 - [What is the REDCap API?](#)
 - ☒ API Export
 - ☒ API Import/Update

Figure 53. “Full Rights” User Role | Other privileges

Settings pertaining to the REDCap Mobile App:

☒ REDCap Mobile App ☐ Allows user to collect data offline in the mobile app
[What is the REDCap Mobile App?](#)

Allow user to download data for all records to the app? ☐

Settings pertaining to project records: [Explain these settings](#)

☒ Create Records ☒
☐ Rename Records ☐
☒ Delete Records ☒
* Includes ability to delete all data on an instrument or on a repeating event.

Settings pertaining to record locking and E-signatures:

☒ Record Locking Customization ☒
☐ Lock/Unlock Records (instrument level) ☐ Disabled ☐ Locking / Unlocking ☒ Locking / Unlocking with E-signature authority
Users with locking privileges also have access to the E-signature and Locking Mgmt page on the left-hand Applications menu.
[What is an E-signature?](#)
[Watch video about locking](#)

☒ Lock/Unlock *Entire* Records (record level) ☒

NOTE: It is important to note that instrument level locking and record level locking are independent features that are governed by separate user privileges (as seen above). You must have explicit permission to either one in order to perform that specific locking action. Also, record locking is a higher-level locking than instrument locking, which means that an entire record may be locked or unlocked while one or more instruments are currently locked, but an instrument cannot be locked or unlocked while the entire record is locked.

Figure 54. "Full Rights" User Role | Other privileges

Privileges for Viewing and Exporting Data								
Data Viewing Rights pertain to a user's ability to view or edit data on pages in the project (e.g., data entry forms, reports). Users with 'No Access' Data Viewing Rights for a given instrument will not be able to view that instrument for any record, nor will they be able to view fields from that instrument on a report. Data Export Rights pertain to a user's ability to export data from the project, whether through the Data Exports page, API, Mobile App, or in PDFs of instruments containing record data. Note: Data Viewing Rights and Data Export Rights are completely separate and do not impact one another.								
	Data Viewing Rights				Data Export Rights			
	No Access (Hidden)	Read Only	View & Edit	Edit survey responses	No Access	De-Identified*	Remove All Identifier Fields	Full Data Set
Pre_Consent	☐	☐	☒		☐	☐	☐	☒
ICF_Survey (survey)	☐	☐	☒	☐	☐	☐	☐	☒
Attest_Survey (survey)	☐	☐	☒	☐	☐	☐	☐	☒
PDF_Uploads	☐	☐	☒		☐	☐	☐	☒

* De-identified means that all free-form text fields will be removed, as well as any date/time fields and Identifier fields.

Figure 55. "Full Rights" User Role | Viewing/Exporting Data

11.2 "Attestation" User Role permissions

The below screenshots provide the permissions for the "Attestation" User Role.

Basic Privileges

Role name:

Highest level privileges:

☐ Project Design and Setup ☐
☒ User Rights ☐ No Access ☒ Read Only ☐ Full Access
☐ Data Access Groups ☐

Figure 56. "Attestation" User Role | Name & Highest-level privileges

Other privileges:

- ☒ Survey Distribution Tools
- ☐ Alerts & Notifications
- ☐ Calendar
- ☒ Add/Edit/Organize Reports
 - Also allows user to view ALL reports (but not necessarily all data in the reports)
- ☒ Stats & Charts
- ☐ Data Import Tool
- ☐ Data Comparison Tool
- ☐ Logging
- ☐ File Repository
- ☒ Data Quality
 - ☐ Create & edit rules
 - ☐ Execute rules
- ☐ Data Resolution Workflow
 - ☐ No Access
 - ☐ View only
 - ☐ Open queries only
 - ☐ Respond only to opened queries
 - ☒ Open and respond to queries
 - ☐ Open, close, and respond to queries
- ☐ API
 - ☐ API Export
 - ☐ API Import/Update

Figure 57. "Attestation" User Role | Other privileges

Settings pertaining to the REDCap Mobile App:

- ☒ REDCap Mobile App
 - ☐ Allows user to collect data offline in the mobile app
- ☐ Allow user to download data for all records to the app?

Settings pertaining to project records: [Explain these settings](#)

- ☒ Create Records
- ☐ Rename Records
- ☐ Delete Records
 - * Includes ability to delete all data on an instrument or on a repeating event.

Settings pertaining to record locking and E-signatures:

- ☐ Record Locking Customization
- ☒ Lock/Unlock Records (instrument level)
 - ☒ Disabled
 - ☐ Locking / Unlocking
 - ☐ Locking / Unlocking with E-signature authority
 - [What is an E-signature?](#)
- ☐ Watch video about locking
- ☐ Lock/Unlock *Entire* Records (record level)

NOTE: It is important to note that instrument level locking and record level locking are independent features that are governed by separate user privileges (as seen above). You must have explicit permission to either one in order to perform that specific locking action. Also, record locking is a higher-level locking than instrument locking, which means that an entire record may be locked or unlocked while one or more instruments are currently locked, but an instrument cannot be locked or unlocked while the entire record is locked.

Figure 58. "Attestation" User Role | Other privileges

Privileges for Viewing and Exporting Data

Data Viewing Rights pertain to a user's ability to view or edit data on pages in the project (e.g., data entry forms, reports). Users with 'No Access' Data Viewing Rights for an instrument will not be able to view that instrument for any record, nor will they be able to view fields from that instrument on a report. Data Export Rights pertain to a user's ability to export data from the project, whether through the Data Exports page, API, Mobile App, or in PDFs of instruments containing record data. Note: Data Viewing Rights and Data Export Rights are completely separate and do not impact one another.

	Data Viewing Rights				Data Export Rights			
	No Access (Hidden)	Read Only	View & Edit	Edit survey responses	No Access	De-identified*	Remove All Identifier Fields	Full Data Set
Pre_Consent	☐	☐	☒		☐	☐	☐	☒
ICF_Survey (survey)	☐	☐	☒	☐	☐	☐	☐	☒
Attest_Survey (survey)	☐	☐	☒	☐	☐	☐	☐	☒
PDF_Uploads	☐	☐	☒		☐	☐	☐	☒

* De-identified means that all free-form text fields will be removed, as well as any date/time fields and Identifier fields.

Figure 59. "Attestation" User Role | Viewing/Exporting Data

OCR Operations Guidance: Using Penn Medicine REDCap for eConsents

12 Revision History

Version	Date	Changes	Changes by:
1	2/19/2024	Document creation	Paul Sanborn